

Muhammad Awaiz

+92 325 0793007 | awaishanif0007@gmail.com | Johar Town, Lahore | [LinkedIn](#) | [GitHub](#)

SUMMARY

Results-driven **AI/ML & Data Engineer** with hands-on experience delivering production-grade **Generative AI, NLP, and Deep Learning** solutions. Proven ability to reduce inference latency by 40%, achieve 92%+ mAP, and deploy scalable AI systems serving 1,000+ daily users. Proficient in building end-to-end pipelines using **Python, PyTorch, TensorFlow, FastAPI**, and **Scikit-learn**, with a consistent focus on measurable business impact.

PROFESSIONAL EXPERIENCE

AI Engineer

Feb '26 — Present

Neurovise AI Solutions

Lahore, Pakistan (Hybrid)

- Architecting and maintaining production-grade AI voice calling systems integrating **OpenAI Realtime API, ElevenLabs**, and **Twilio** with sub-300ms end-to-end latency
- Building and optimizing real-time conversational AI pipelines with WebSocket streaming, connection pooling, and multilingual support (English/Urdu)
- Integrating **MongoDB** and Google Calendar API for lead management and automated meeting scheduling

Data Engineer & AI Engineer

May '25 — Feb '26

BiolCAW Technologies

Rawalpindi, Pakistan

- Led end-to-end AI engineering projects from data collection and preprocessing to model deployment and production maintenance, ensuring scalable and reliable AI solutions
- Developed and deployed **computer vision** and **NLP** applications achieving sub-50ms inference latency for real-time industrial use cases
- Built production-ready **ML pipelines** with **FastAPI** and **Docker** containerization, implementing **CI/CD** workflows for automated deployment
- Designed data collection frameworks including web scraping, API integration, and automated annotation pipelines
- Performed model optimization through quantization and **GPU acceleration** using **CUDA**, reducing inference time by 40%
- Implemented **MLOps** practices with **MLflow** for experiment tracking, model versioning, and automated retraining pipelines

Data Science Fellow

May '24 — Aug '24

Bytewise Limited

Remote

- Built classification and regression models achieving 90%+ accuracy using **Scikit-learn** and **XGBoost**
- Developed customer churn prediction system with advanced feature engineering reducing churn by 25%
- Conducted comprehensive data analysis and created interactive dashboards for actionable business insights

PROJECTS

AI Outbound Voice Calling Agent — Real-Time Conversational AI System

- Architected production-grade AI calling agent integrating **OpenAI Realtime API** for live STT and LLM-driven conversation with sub-300ms end-to-end latency
- Engineered hybrid TTS pipeline using **ElevenLabs** voice cloning with WebSocket streaming and HTTP fallback; integrated **Twilio, MongoDB**, and Google Calendar API for call orchestration and multilingual support

KYC Verification Platform — End-to-End Identity & Compliance System

- Built production-grade KYC API using **FastAPI** supporting document verification (CNIC, Passport, Driving License), liveness detection, and 1:1 face matching with sub-50ms inference latency
- Integrated AML screening pipeline and **Stripe** subscription system with webhook-driven lifecycle management and tiered usage quotas tracked via **MongoDB**

Real-Time Helmet Detection System Using YOLOv8

- Trained **YOLOv8** model on 3,500+ annotated images achieving 92.3% mAP; deployed FastAPI inference API with <45ms latency

- Implemented GPU-accelerated pipeline with **CUDA** reducing inference time by 40%; integrated with live camera feeds at 30+ FPS

AI Medical Assistant — RAG-Powered Diagnostic Support System

- Built RAG system processing 10,000+ medical encyclopedia entries with **FAISS**; achieved 93% context relevance using Hugging Face embeddings
- Integrated **Groq API LLaMA-70B** and GPT models for context-aware retrieval; developed Streamlit interface serving 1,000+ daily queries

OratoAI — AI-Powered Public Speaking Coach (FYP)

- Architected multi-modal **LangGraph** agentic AI system orchestrating specialized agents across audio, visual, and NLP pipelines for holistic public speaking feedback
- Delivered composite 0–10 score with actionable recommendations via full-stack dashboard built with **FastAPI** backend and modern frontend

SKILLS

Languages: Python, SQL

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, FastAPI, LangChain, FAISS, Streamlit

Computer Vision & Audio AI: YOLOv8, VGG16, OpenCV, Whisper, ElevenLabs

AI/ML Tools: MLflow, Hugging Face, OpenAI API, Groq API

Data Engineering: Scrapy, Selenium, MongoDB, Docker, REST APIs, Twilio, Stripe, Google Calendar API

DevOps & MLOps: Docker, CI/CD, CUDA, Git, MLflow

EDUCATION

BS in Computer Science

Jul '25

Pir Mehr Ali Shah Arid Agriculture University

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision